TIM-LOUIS WILKEN

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Profile: quantitative forecasting and optimisation for power markets; Python scripting, SQL, R.

EDUCATION

UNIVERSITY COLLEGE LONDON (UCL) / UNIVERSITY OF OXFORD — London / Oxford, United Kingdom
M.Sc. Energy Systems and Data Analytics (UCL); thesis as visiting student (Oxford) — *Sep 2025 – Sep 2026*

- **Average 76.8/100** across all taught modules (Distinction ≥ 70; dissertation pending). Top marks: Statistics for Energy Analytics **89**, Energy and Transport Analytics **85**, Econometrics of Energy Markets **82**.
- Thesis, written at Oxford (Jun–Sep 2026, fee waiver), supervised by A. Stratigakos (UCL), T. Morstyn and Y. Zhou (Oxford): *Deadline Commitments for GPU Jobs under a Forecast-Uncertain Non-Firm Grid Connection*. Two-stage stochastic MILP with ECMWF weather ensembles and chance constraints, hedging firm commitments against volumetric uncertainty; 6–21 pp more deadlines than a greedy scheduler.

TECHNICAL UNIVERSITY OF MUNICH (TUM) — Munich, Germany
B.Sc. Management and Technology — *Oct 2021 – Mar 2025*

- Final grade 2.2 (German scale, 1.0 = best); electrical engineering, business analytics and finance; exchange semester at Singapore Management University (2023).
- Thesis (grade 1.3), *Rising connection requests for BESS in Germany*: analysed the 2024 price volatility, renewable expansion and battery cost degression behind 340 GW of BESS grid-connection requests, quantifying arbitrage revenues.

SIMON-MARIUS-GYMNASIUM — Gunzenhausen, Germany
Abitur; DPG-Abiturpreis for outstanding achievement in physics — *2012 – 2020*

TECHNICAL PROJECTS

- **GB Day-Ahead Price Forecasting**: online ensemble of 13 models (LSTM, Transformer, gradient boosting, LEAR and others) on demand, wind, solar, gas and carbon inputs, retrained daily to track changing market conditions and combined via Bernstein Online Aggregation; 21 % lower MAE than the LEAR benchmark over 2024–2026 (Diebold–Mariano $p < 0.001$); calibrated probabilistic forecasts; in an algorithmic day-ahead battery dispatch backtest (50 MW / 100 MWh) it captures 93.7 % of the perfect-foresight arbitrage revenue in 2026 (93.1 % in 2025), vs. 90.4 % for LEAR and 76.5 % for persistence; strict leakage audit (pre-auction data only). *(Python, PyTorch, scikit-learn; live daily at* timlouiswilken.github.io/gb-dayahead-forecast*)*
- **BESS Trading Simulator**: web app for GB battery trading across day-ahead, intraday and imbalance markets on live Elexon BMRS data, with balancing-mechanism acceptances, a risk book and a P&L dashboard. *(React, TypeScript; built with a fellow student, 2026;* bess-trading-simulator.com*)*
- **Australian Electricity-Market Dynamics**: three-regime threshold vector error-correction (TVECM) analysis of how battery storage shapes SA–VIC wholesale price spillovers (52,608 half-hourly observations, 2SLS). *(Python, statsmodels;* github.com/TimLouisWilken/analysis-of-the-australian-electricity-market*)*

PROFESSIONAL EXPERIENCE

JOH. BERENBERG, GOSSLER & CO. KG — Munich, Germany
Origination and Growth Intern (full-time) — *May 2025 – Aug 2025*

- Built data-driven lead-generation workflows across 5 client segments, scaling to ∼100 qualified leads per week.

LUX-WERFT UND SCHIFFFAHRT GMBH — Pleinfeld, Germany
Ship Operator, MS Brombachsee, Europe's largest inland passenger trimaran (part-time) — *Jun 2024 – Aug 2025*

MOONROC ADVISORY PARTNERS GMBH — Munich, Germany
Strategy Consulting Intern (full-time) — *Feb 2024 – May 2024*

- Quantified interview data for strategy workshops; co-authored *Retail Banking Compass 2024* (fintech chapter).

KPMG AG (AUDITING & ADVISORY) — Munich, Germany
Research Assistant, Audit (full-time) — *May 2023 – Aug 2023*

- Developed a client toolkit for sustainability reporting with TUM's Chair of Financial Accounting.

WEBSHAPE WEBDESIGN — Munich, Germany
Co-Founder & Managing Director (self-employed) — *Mar 2020 – Oct 2022*

- Co-founded a profitable web-design agency with high five-figure revenue, delivering 35+ client projects.

AWARDS & LEADERSHIP

- **Rolls-Royce University Project** (2025–26): reactor systems design (HTGR); team awarded first place.
- **Cambridge Climate Challenge** (2026): team lead of a three-person interdisciplinary team; finalist.
- **UCL Energy Society** (2025–26): External Speaker Officer, sourcing industry experts for panels and keynotes.

SKILLS

Energy markets	Day-ahead, intraday and imbalance markets (GB, DE); BESS arbitrage; Elexon and ECMWF data.
Methods	Statistical modelling, forecasting; optimisation under uncertainty; risk management; econometrics.
Tools	Python (pandas, PyTorch, scikit-learn, statsmodels, PuLP); HiGHS, Gurobi; SQL, R, Git, TypeScript.
Languages	German (native), English (C1).